

Chemistry Ch 2.3: Allotropes of Carbon

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Allotropy

Kisi element ki wo khasoosiyat ke wo **alag alag physical shaklon** mein mojood ho sakta hai. Yani atom wohi hain, sirf **jurne ka andaz** alag hai, aur isi se cheez bilkul badal jati hai.

Shakal	Banawat	Properties
Graphite	Do rukhi (2D) hexagonal tehen. Tehen ke darmiyan bond kamzor .	Naram aur phisalna , aur bijli ka acha conductor
Diamond	Har carbon chaar doosre carbon se jura, tetrahedral shakal.	Sab se sakht maddah, melting point bohat ooncha , bijli nahin chalata
Buckyballs (C-60, Fullerenes)	Football jaisi khokhli gaind: 20 hexagons + 12 pentagons , 60 carbon atoms, har ek teen se jura.	Khokhla aur gol

Do sawal jo hamesha aate hain

Graphite naram kyun hai jab carbon itna mazboot hai? Kyunke tehen ke **andar** bond mazboot hain magar **tehon ke darmiyan** kamzor. Is liye tehen ek doosri par phisal jati hain. Pencil isi tarah likhti hai.
Diamond bijli kyun nahin chalata? Kyunke har carbon apne **chaaron** electrons bond mein laga chuka hai, to koi **free electron** bacha hi nahin. Graphite mein chautha electron azad hai, isi liye wo chalata hai.

Number yaad rakhein

Buckyball: **60** carbon atoms, **20** hexagons, **12** pentagons, har atom **3** se jura. Diamond mein har atom **4** se jura. Ye 3 aur 4 ka farq hi sab kuch badal deta hai.

Quiz

1. Define **allotropy**, then name the three allotropes of carbon your book gives and describe the structure of each.
2. Graphite and diamond are both pure carbon. Explain why graphite is soft and conducts electricity while diamond is the hardest substance and does not conduct.